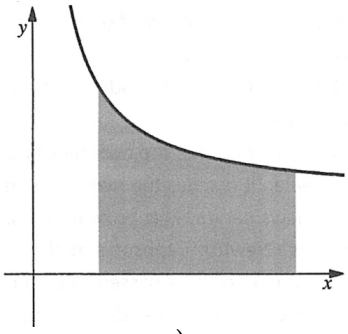
Integration

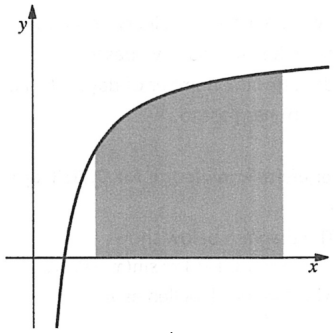
Questions 1–264

2.30 — Question 1. The four shaded areas below are to be estimated using the trapezium rule. Which would produce an underestimate?

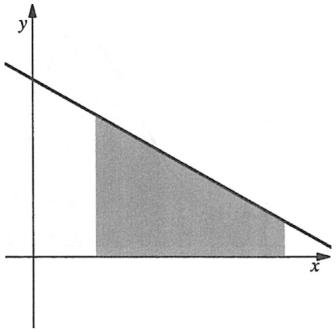
a)



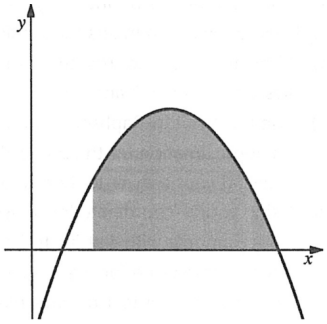
b)



c)



d)



2.30 — Question 2. Amelia uses the trapezium rule to estimate an area from $x = 3$ to $x = 8$ using 9 ordinates. What is the horizontal width of each trapezium?

2.30 — Question 3. Use the trapezium rule, with the stated number of strips, to estimate the following integrals. Give your answers in exact form.

a) $\int_0^3 x^2\,dx$, 3 strips

b) $\int_0^4 (x^2+1)\,dx$, 4 strips

c) $\int_0^2 x^3\,dx$, 4 strips

d) $\int_0^3 (x^3+2)\,dx$, 6 strips

2.30 — Question 4. Use the trapezium rule, with the stated number of ordinates, to estimate the following integrals. Give your answers in exact form.

a) $\int_2^4 (2x^2+3x)\,dx$, 5 ordinates

b) $\int_1^2 (4-(x-1)^2)\,dx$, 5 ordinates

c) $\int_1^4 (20-x^2)\,dx$, 7 ordinates

d) $\int_{-3}^2 ((x+2)^2+3)\,dx$, 6 ordinates

2.30 — Question 5. Use the trapezium rule, with the stated number of strips, to estimate the following integrals. Give your answers in exact form.

a) $\int_{-3}^{1}(x^3+2x^2+10)\,dx$, 4 strips

b) $\int_{-2}^{2}(5-(x-1)(x+2))\,dx$, 8 strips

c) $\int_{0}^{2}(3-\frac{1}{2}x^2)\,dx$, 5 strips

d) $\int_{-3}^{1}(2-\frac{1}{3}x^3)\,dx$, 5 strips

2.30 — Question 6. Use the trapezium rule, with the stated number of ordinates, to estimate the following integrals. Give your answers in exact form.

a) $\int_{2}^{8}(8-x)(x-1)(x+2)\,dx$, 5 ordinates

b) $\int_{1}^{9}(3^x+2)\,dx$, 5 ordinates

c) $\int_{-7}^{-4}(\frac{2}{x+2}+2)\,dx$, 7 ordinates

d) $\int_{5}^{7}\frac{x+2}{x-4}\,dx$, 4 ordinates

2.30 — Question 7. Use the trapezium rule, with four strips, to find an estimate for $\int_1^2 \frac{\ln x}{\sqrt{e^x+1}}\,dx$

2.30 — Question 8. Given that $k > 0$, use the trapezium rule, with four strips, to show that:

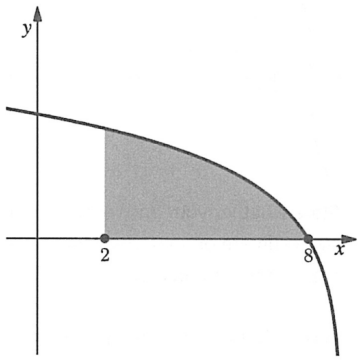
a) $\int_0^8 \frac{k+x^2}{2}\,dx = 4k + 88$

b) $\int_1^5 \frac{k+x}{x}\,dx = \frac{101k+240}{60}$

c) $\int_1^5 \ln(kx)\,dx = \ln(24\sqrt{5}k^4)$

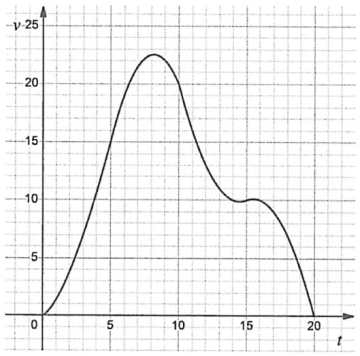
d) $\int_1^5 \ln(\frac{x}{k})\,dx = \ln 24 + \frac{1}{2}\ln 5 - 4\ln k$

2.30 — Question 9. The curve shown below has the equation $y = \ln(9 - x)$.



Use three trapeziums to estimate the shaded region, giving your answer to three decimal places. Explain how you know this will give an underestimate.

2.30 — Question 10. A particle is moving in a straight line with velocity v ms^{-1} at time t seconds as shown in the graph below. Use the trapezium rule with four strips to estimate the distance travelled by the particle in the first 20 seconds.



2.30 — Question 11.

a) Use the trapezium rule, with four strips, to find an estimate for $\int_0^1 \frac{1}{\sqrt{4+x^2}}\,dx$

b) Explain how the trapezium rule might be used to give a better estimate to the integral given in a).

2.30 — Question 12. Use the trapezium rule, with six ordinates, to find an estimate for $\int_0^1 \frac{x}{\sqrt{x^3+1}}\,dx$

2.30 — Question 13. A particle is moving in a straight line and its speed is measured every 3 seconds. The results are given in the table below with the time in seconds and the speed in ms^{-1}. Use this information with the trapezium rule to estimate the distance travelled by the particle.

Time (s)	0	3	6	9	12	15	18
Speed (ms^{-1})	1	2.8	4.4	6.2	7.7	8.9	10.1

2.30 — Question 14.

a) The table below shows the values of x and y for $y = \sqrt{\frac{2+x}{x}}$

x	2.5	3	3.5	4	4.5
y	1.342	1.291	1.254	1.225	1.202

The values of y are given to 4 significant figures. Use the trapezium rule with all the values of y in the table to estimate $\int_{2.5}^{4.5} \sqrt{\frac{2+x}{x}}\, dx$

b) Use your answer to deduce an estimate to $\int_{2.5}^{4.5} \sqrt{\frac{2+x}{2x}}\, dx$ giving your answer to 3 significant figures.

2.30 — Question 15.

a) Use the trapezium rule, with four strips, to find an estimate for $\int_1^2 e^{3x^2}\,dx$

b) Use your answer to deduce an estimate to $\int_1^2 e^{3x^2+1}\,dx$

2.30 — Question 16. The screenshot below shows part of a calculation to use the trapezium with five strips to estimate $\int_0^2 e^{8-x^3}\,dx$

	A	B	C	D
1	x	8-x^3	y	
2	0.0000	8.0000	2980.9580	1490.4790
3	0.4000	7.9360	2796.1535	2796.1535
4	0.8000	7.4880	1786.4756	1786.4756
5	1.2000	6.2720	529.5354	529.5354
6	1.6000	3.9040	49.6005	49.6005
7	2.0000	0.0000	1.0000	0.5000
8				6652.7439

Use the values in the spreadsheet to estimate the value of the integral to 3 significant figures.

2.30 — Question 17.

a) The table below shows the values of x and y for $y = \log_5 \frac{x}{6}$

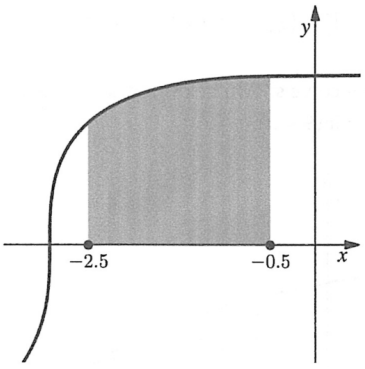
x	8	12	16	20	24
y	0.1787	0.4307	0.6094	0.7481	0.8614

The values of y are given to 4 decimal places. Use the trapezium rule with all the values of y in the table to estimate $\int_8^{24} \log_5 \frac{x}{6}\, dx$

b) Use your answer to deduce an estimate to $\int_8^{24} \log_5 \frac{x^2}{36}\, dx$

2.30 — Question 18.

a) The diagram below shows the curve with equation $y = \sqrt[3]{x^3 + 25}$. Use the trapezium rule with five ordinates to estimate $\int_{-2.5}^{-0.5} \sqrt[3]{x^3 + 25}\, dx$



b) State whether your answer to a) is an overestimate or an underestimate, giving a reason for your answer.

2.30 — Question 19. The screenshot below shows part of a calculation to use the trapezium with five strips to estimate $\int_0^6 \sqrt{e^{-x} - e^{-2x}}\, dx$

	A	B	C	D	E
1	x	exp(-x)	exp(-2x)	y	
2	0.0000	1.0000	1.0000	0.0000	0.0000
3	1.2000	0.3012	0.0907	0.4588	0.4588
4	2.4000	0.0907	0.0082	0.2872	0.2872
5	3.6000	0.0273	0.0007	0.1630	0.1630
6	4.8000	0.0082	0.0001	0.0903	0.0903
7	6.0000	0.0025	0.0000	0.0497	0.0249
8					1.0242

Use the values in the spreadsheet to estimate the value of the integral to 3 significant figures.

2.30 — Question 20. Prerequisite knowledge for Q20: Radians Use the trapezium rule, with six ordinates, to find an estimate for $\int_0^{\pi} \sin^2 x\, dx$

2.30 — Question 21. The table below shows the values of x and y for $y = \mathrm{f}(x)$

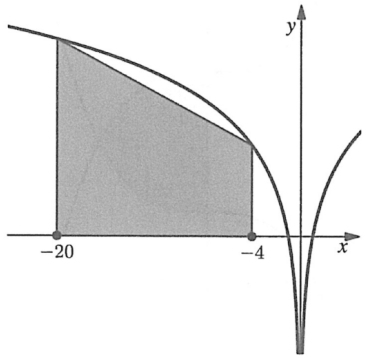
x	5	5.4	5.8	6.2	6.6
y	p	15	22	p	18

a) Use the trapezium rule with all the values of y in the table to estimate $\int_5^{6.6} \mathrm{f}(x)\,dx$ in terms of p

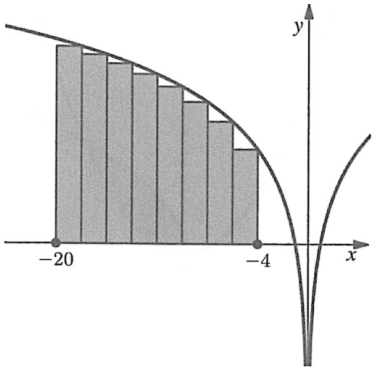
b) It is given that the estimate in part a) is 25. Find the value of p

2.30 — Question 22. The diagram below shows the curve with equation $y = \ln(x^2)$

a) The area bounded by the curve, the x-axis, and the lines $x = -20$ and $x = -4$, is approximated by a single trapezium. Show that the area of the trapezium is $64\ln 2 + 16\ln 5$

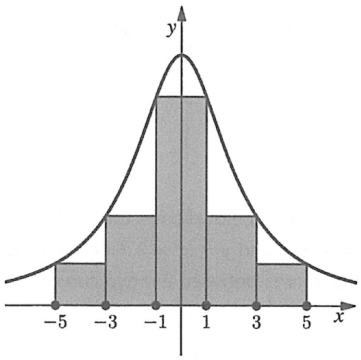


b) The area is now approximated using eight rectangles of equal width. Show that the total area of the rectangles is $60\ln 2 + 16\ln 3 + 4\ln 5 + 4\ln 7$

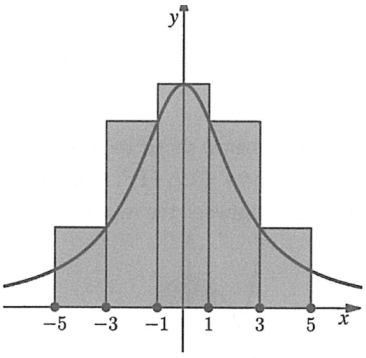


2.30 — Question 23. Prerequisite knowledge: Integration by Substitution (for part d) Laura is trying to find an estimate for $\int_{-5}^{5} \frac{25}{5+x^2}\,dx$

a) She first uses five rectangles of equal width as shown in the diagram below. Show that this lower bound for the area is given by18.81 to 4 significant figures.



b) Laura then finds an upper bound for the value of the integral using five rectangles of equal width as shown in the diagram below. Determine the value of this upper bound to 4 significant figures.

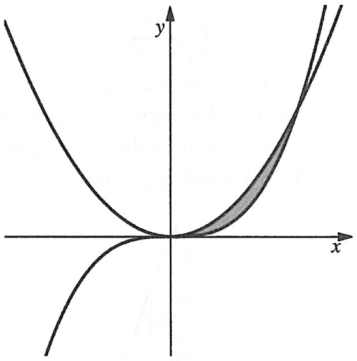


2.30 — Question 23 (continued). Prerequisite knowledge: Integration by Substitution (for part d) Laura is trying to find an estimate for $\int_{-5}^{5} \frac{25}{5+x^2}\,dx$

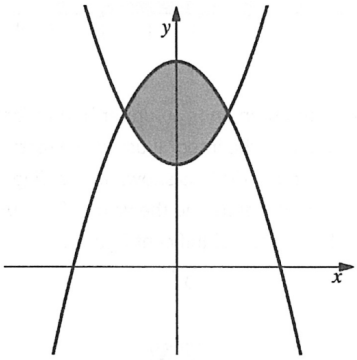
c) Laura then uses the trapezium rule with five strips to estimate the value of $\int_{-5}^{5} \frac{25}{5+x^2}\,dx$. Find her result to 4 significant figures.

d) Finally, Laura uses the substitution $x = \sqrt{5}\tan u$ to integrate and find an accurate value for the area. Find her result to 4 significant figures.

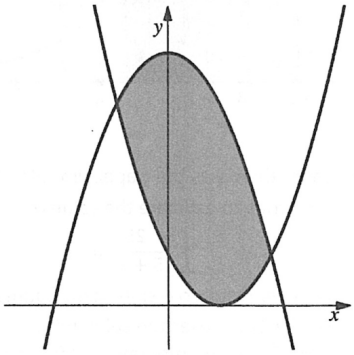
2.31 — Question 1. The diagram below shows the curves $y = x^2$ and $y = x^3$. Find the exact shaded area between the two curves.



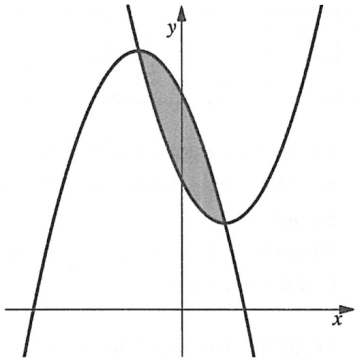
2.31 — Question 2. The diagram below shows the curves $y = 4 - x^2$ and $y = x^2 + 2$. Find the exact shaded area between the two curves.



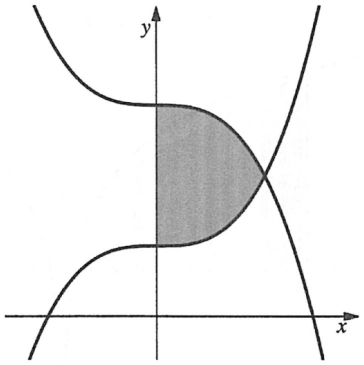
2.31 — Question 3. The diagram below shows the curves $y = 5 - x^2$ and $y = (x-1)^2$. Find the exact shaded area between the two curves.



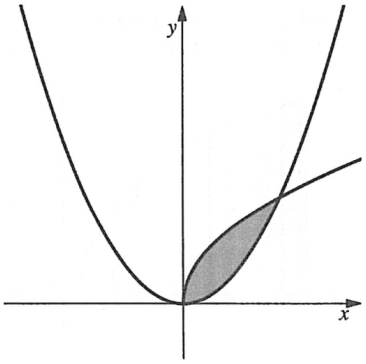
2.31 — Question 4. The diagram below shows the curves $y = (x-1)^2 + 2$ and $y = 6 - (x+1)^2$. Find the exact shaded area between the two curves.



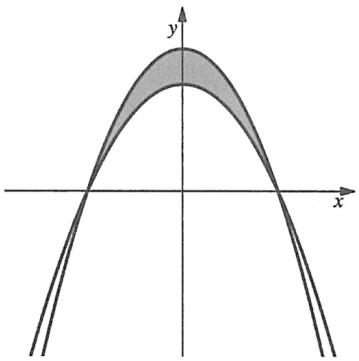
2.31 — Question 5. The diagram below shows the curves $y = x^3 + 1$ and $y = 3 - x^3$. Find the exact shaded area between the two curves and the y-axis.



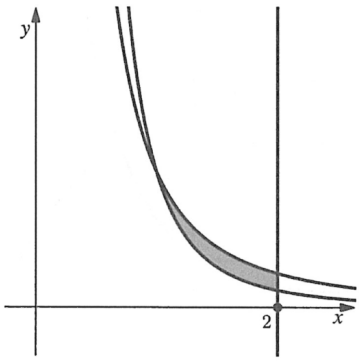
2.31 — Question 6. The diagram below shows the curves $y = x^2$ and $y = \sqrt{x}$. Find the exact shaded area between the two curves.



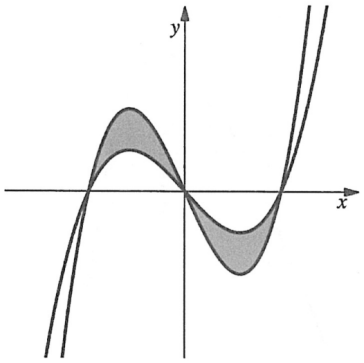
2.31 — Question 7. The diagram below shows the curves $y = 1 - x^2$ and $y = \frac{4}{3}(1 - x^2)$. Find the exact shaded area between the two curves.



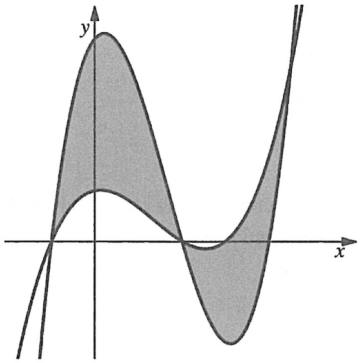
2.31 — Question 8. The diagram below shows the curves $y = \frac{1}{x^2}$ and $y = \frac{1}{x^3}$, and the line $x = 2$. Find the exact shaded area.



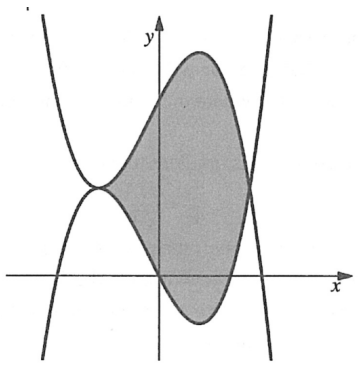
2.31 — Question 9. The diagram below shows the curves $y = x(x-1)(x+1)$ and $y = 2x(x-1)(x+1)$. Find the exact shaded area between the two curves.



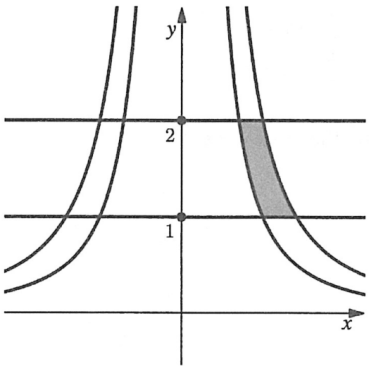
2.31 — Question 10. The diagram below shows the curves $y = (x+1)(x-2)(x-3)$ and $y = 3(x+1)(x-2)(x-4)$. Find the exact shaded area.



2.31 — Question 11. It is given that $\mathrm{f}(x) = 8x^3 + 6x^2 - 36x$. The diagram below shows the curves $y = \mathrm{f}(x)$ and $y = 81 - \mathrm{f}(x)$. The curves intersect at precisely two points. Find the exact shaded area.



2.31 — Question 12. The diagram below shows the curves $y = \frac{1}{x^2}$ and $y = \frac{2}{x^2}$, and the lines $y = 1$ and $y = 2$. Find the exact shaded area.



2.32 — Question 1.

Integration of e^{kx} Find

a) $\int (e^x + x)\, dx$

b) $\int e^{2x}\, dx$

c) $\int (2e^{3x} - x^2)\, dx$

d) $\int (\sqrt{e^x} - \frac{9}{x^2})\, dx$

e) $\int \frac{2 + e^x}{e^{8x}}\, dx$

2.32 — Question 2. Find the equation of the curve y given the gradient function is:

a) $\frac{dy}{dx} = e^{-3x}$

b) $\frac{dy}{dx} = 4 - 2e^{-2x}$

c) $\frac{dy}{dx} = \sqrt{x} - \frac{1}{3}e^{x/6}$

d) $\frac{dy}{dx} = x - \frac{6}{e^{4x}}$

2.32 — Question 3. The curve y has gradient function $\dfrac{dy}{dx} = 12e^{6x}$ and passes through the point $(\ln 2, 128)$. Show that y can be written in the form $y = ae^{bx} + c$, where a, b and c are integers.

2.32 — Question 4. The curve y has gradient function $\dfrac{dy}{dx} = \frac{24}{\sqrt{e^{3x}}}$ and passes through the point $(\ln 4, 1)$. Show that y can be written in the form $y = a + be^{cx}$, where a and b are integers and c is a rational number.

2.32 — Question 5. Evaluate the following definite integrals:

a) $\int_0^3 (e^x + 2)dx$

b) $\int_{\ln 2}^{\ln 3} (100 - e^x)dx$

c) $\int_{-1}^{1} (24e^{2x/3} - 2x)dx$

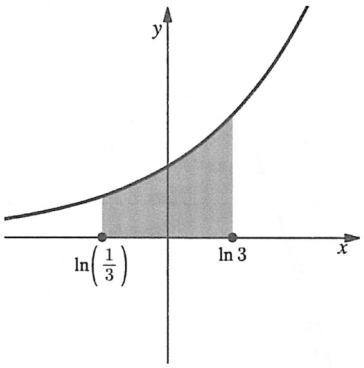
d) $\int_0^{\ln 2} (\frac{5}{e^x} - \frac{2}{e^{2x}})dx$

2.32 — Question 6. Find the exact value of:

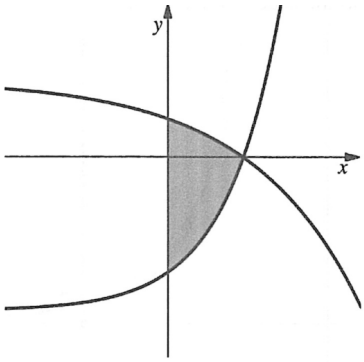
a) $\lim_{\delta x \to 0} \sum_{x=0}^{1} 3e^{18x}\delta x$

b) $\lim_{\delta x \to 0} \sum_{x=2}^{\ln 8} \frac{1}{\sqrt[3]{e^x}}\delta x$

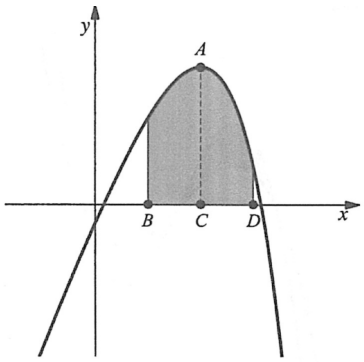
2.32 — Question 7. The diagram below shows the curve with equation $y = \sqrt{e^x}$. Find the exact value of the shaded area.



2.32 — Question 8. The diagram below shows the curves $y = 2 - e^x$ and $y = e^{2x} - 4$. Find the exact shaded area between the two curves.



2.32 — Question 9. The diagram below shows the curve with equation $y = e^2x - e^x$. The curve has a turning point at A, and AC bisects BD. It is given that $BD = 2$ units. Find the exact shaded area.



2.32 — Question 10. Integration of $\sin kx$ and $\cos kx$ Find

a) $\int (\sin x + \cos x)\, dx$

b) $\int (2\sin x - \cos x)\, dx$

c) $\int (2\sin 2x + \cos 4x)\, dx$

d) $\int (3\sin \frac{1}{2}x - 6\cos \frac{2}{3}x)\, dx$

2.32 — Question 11. Find the equation of the curve y given the gradient function is:

a) $\dfrac{dy}{dx} = 3\cos 6x$

b) $\dfrac{dy}{dx} = \frac{3}{x^2} - 4\sin\frac{1}{4}x$

c) $\dfrac{dy}{dx} = \sqrt[3]{x} - \frac{2}{3}\cos\frac{5}{6}x$

2.32 — Question 12. The curve y has gradient function $\dfrac{dy}{dx} = -4\sin 3x$ and passes through the point $(\frac{\pi}{6}, 3)$. Show that y can be written in the form $y = a\cos bx + c$, where b and c are integers, and a is a rational number.

2.32 — Question 13. The curve y has gradient function $\dfrac{dy}{dx} = 2\sin\frac{1}{3}x - 3\cos\frac{1}{3}x$ and passes through the point $(\pi, -\frac{9\sqrt{3}}{2} - 3)$. Show that y can be written in the form $y = a\cos\frac{x}{b} + c\sin\frac{x}{b}$, where a, b and c are integers.

2.32 — Question 14. Evaluate the following definite integrals:

a) $\displaystyle\int_0^{\pi/2} \sin x\, dx$

b) $\displaystyle\int_0^{\pi/4} \cos 2x\, dx$

c) $\displaystyle\int_{-5\pi/6}^{5\pi/3} 3\sin\frac{1}{5}x\, dx$

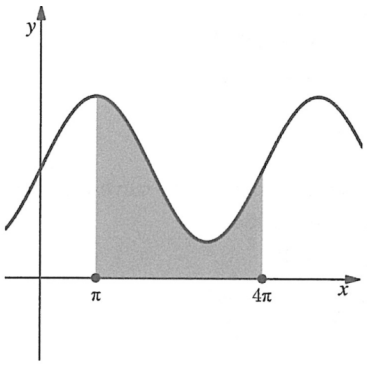
d) $\displaystyle\int_{\pi/8}^{\pi/4} (5\sin 4x - 6\cos 4x)dx$

2.32 — Question 15. Find the exact value of:

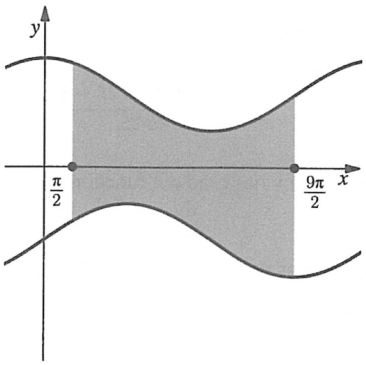
a) $\lim_{\delta x \to 0} \sum_{x=0}^{\pi/12} (8 \sin 4x - 16 \cos 4x) \delta x$

b) $\lim_{\delta x \to 0} \sum_{x=9\pi}^{10\pi} (\frac{1}{8} \sin \frac{1}{4} x - \frac{1}{16} \cos \frac{1}{4} x) \delta x$

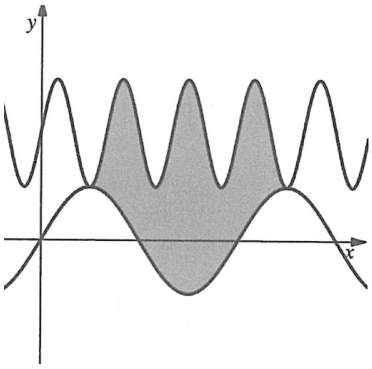
2.32 — Question 16. The diagram below shows the curve with equation $y = 3 + 2 \sin \frac{1}{2} x$. Find the exact value of the shaded area.



2.32 — Question 17. The diagram below shows the curves $y = \sin\frac{1}{3}x - 2$ and $y = \cos\frac{1}{3}x + 2$. Find the exact shaded area between the two curves.



2.32 — Question 18. The diagram below shows the curve with equation $y = \sin x$ and the curve with equation $y = a + \sin bx$, where x is in radians. Find the value of a and the value of b, and hence find the exact shaded area.



2.32 — Question 19. Integration of a^x Find

a) $\int 3^x \, dx$

b) $\int (3^{2x} + 4^{3x}) \, dx$

c) $\int (\frac{1}{5^x} - \frac{2}{2^x}) \, dx$

d) $\int \frac{3^x - 4}{9^x} \, dx$

2.32 — Question 20. Find the equation of the curve y given the gradient function is:

a) $\frac{dy}{dx} = 8^x$

b) $\frac{dy}{dx} = x - 3^x \ln 3$

c) $\frac{dy}{dx} = 6^{3x} - \frac{1}{6^{3x}}$

d) $\frac{dy}{dx} = \frac{16^x - 4}{2^x}$

2.32 — Question 21. The curve y has gradient function $\frac{dy}{dx} = 25^x$ and passes through the point $(0, 20)$. Show that $y = \frac{25^x + 20\ln 25 - 1}{\ln 25}$.

2.32 — Question 22. The curve y has gradient function $\frac{dy}{dx} = \frac{1}{4^x}$ and passes through the point $(0, \frac{1}{\ln 4})$. Show that $y = \frac{1 - 2^{-2x-1}}{\ln 2}$.

2.32 — Question 23. Evaluate the following definite integrals:

a) $\int_{-1}^{1} 3^x dx$

b) $\int_{0}^{2} (2^x - 4^x) dx$

c) $\int_{-1/3}^{1/3} (5^{3x} - x) dx$

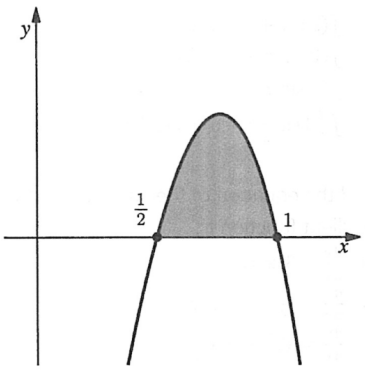
d) $\int_{1}^{2} \frac{2 - 4^x}{4^x} dx$

2.32 — Question 24. Find the exact value of:

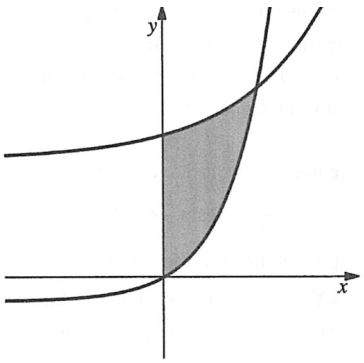
a) $\lim_{\delta x \to 0} \sum_{x=5}^{10} 5^{0.2x} \delta x$

b) $\lim_{\delta x \to 0} \sum_{x=1/10}^{1/5} \frac{10}{2^{5x}} \delta x$

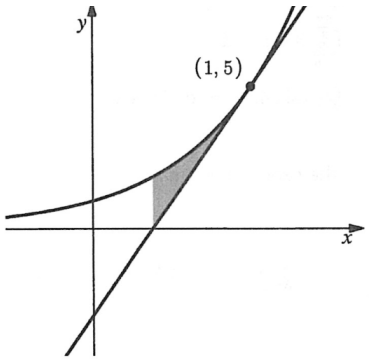
2.32 — Question 25. The diagram below shows the curve with equation $y = 4x - 4^x$. Find the exact value of the shaded area.



2.32 — Question 26. The diagram below shows the curves $y = 2^x + 5$ and $y = 4^x - 1$. Find the exact shaded area between the two curves.



2.32 — Question 27. The diagram below shows the curve with equation $y = 5^x$. The tangent to the curve at $(1, 5)$ is also shown. Find the exact shaded area.



2.32 — Question 28. Integration of $\frac{1}{x}$ Find

a) $\int \frac{1}{x}\,dx$

b) $\int (\frac{3}{x} - \frac{x}{3})\,dx$

c) $\int (\frac{5}{3x} + \frac{2}{5x^2})\,dx$

d) $\int \frac{3+8x}{4x}\,dx$

2.32 — Question 29. Find the equation of the curve y given the gradient function is:

a) $\frac{dy}{dx} = \frac{4}{x} - \frac{4}{x^2}$

b) $\frac{dy}{dx} = 2 - \frac{7}{4x}$

c) $\frac{dy}{dx} = \frac{3\sqrt{x}+1}{x^{3/2}}$

d) $\frac{dy}{dx} = \frac{8x-3}{5x^2}$

2.32 — Question 30. The curve y has gradient function $\frac{dy}{dx} = \frac{5}{x}$ and passes through the point $(e^5, 1)$. Show that y can be written in the form $y = a + b\ln|x|$, where a and b are integers.

2.32 — Question 31. The curve y has gradient function $\frac{dy}{dx} = \frac{3-10x}{x}$ and passes through the point $(1, 1)$. Show that y can be written in the form $y = ax + b + c\ln|x|$, where a, b and c are integers.

2.32 — Question 32. Evaluate the following definite integrals:

a) $\int_1^2 \frac{9}{x} dx$

b) $\int_{-3}^{-1} \frac{8}{x} dx$

c) $\int_1^4 \frac{2+x}{10x} dx$

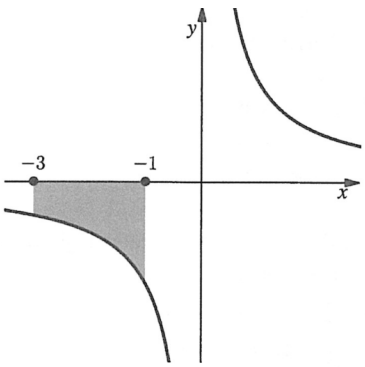
d) $\int_{-3}^{-2} \frac{3-x}{5x} dx$

2.32 — Question 33. Find the exact value of:

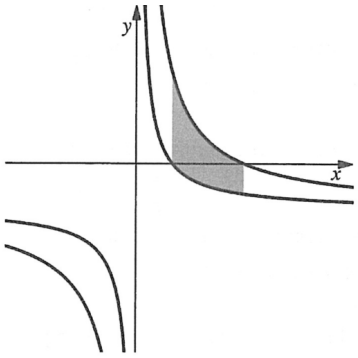
a) $\lim_{\delta x \to 0} \sum_{x=-8}^{-4} (\frac{4}{x} - 1)\delta x$

b) $\lim_{\delta x \to 0} \sum_{x=e}^{e^2} (\frac{3}{2x} + 12x)\delta x$

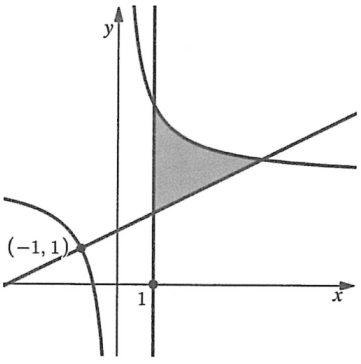
2.32 — Question 34. The diagram below shows the curve with equation $y = \frac{10}{x}$. Find the exact value of the shaded area.



2.32 — Question 35. The diagram below shows the curves $y = \dfrac{9-3x}{x}$ and $y = \dfrac{3-3x}{x}$. Find the exact shaded area between the two curves.



2.32 — Question 36. The diagram below shows the curve with equation $y = 3 + \frac{2}{x}$. The normal to the curve at $(-1, 1)$ is also shown. Find the exact shaded area, bounded by the curve, the normal and the line $x = 1$.



2.32 — Question 37. Integration of Trigonometric Functions Find

a) $\int \sec^2 x \, dx$

b) $\int \operatorname{cosec} x \cot x \, dx$

c) $\int \sec x \tan x \, dx$

d) $\int \operatorname{cosec}^2 x \, dx$

e) $\int \sec^2 4x \, dx$

f) $\int \operatorname{cosec}^2 \frac{1}{3}x \, dx$

2.32 — Question 38. Find the equation of the curve y given the gradient function is:

a) $\frac{dy}{dx} = \operatorname{cosec} 2x \cot 2x$

b) $\frac{dy}{dx} = 8 \operatorname{cosec} \frac{1}{4}x \cot \frac{1}{4}x$

c) $\frac{dy}{dx} = \frac{2}{3} \sec \frac{10}{3}x \tan \frac{10}{3}x$

2.32 — Question 39. The curve y has gradient function $\dfrac{dy}{dx} = 5\sec^2 3x$ and passes through the point $(\frac{\pi}{12}, 5)$. Write y in terms of x.

2.32 — Question 40. The curve y has gradient function $\dfrac{dy}{dx} = \sec \frac{1}{2}x \tan \frac{1}{2}x$ and passes through the point $(\frac{\pi}{3}, 0)$. Write y in terms of x.

2.32 — Question 41. Evaluate the following definite integrals:

a) $\displaystyle\int_0^{\pi/6} 4\sec^2 2x\,dx$

b) $\displaystyle\int_{\pi/2}^{3\pi/2} 9\operatorname{cosec}\frac{1}{3}x\cot\frac{1}{3}x\,dx$

c) $\displaystyle\int_{\pi/6}^{\pi/4} \frac{1}{3}\operatorname{cosec}^2 3x\,dx$

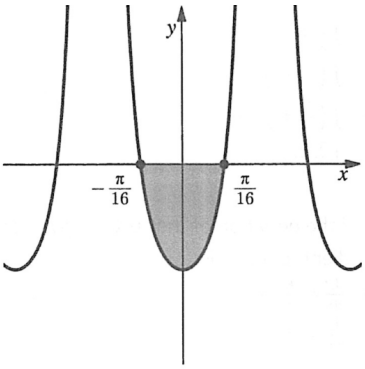
d) $\displaystyle\int_{-\pi/4}^{0} -\sec\frac{2}{3}x\tan\frac{2}{3}x\,dx$

2.32 — Question 42. Find the exact value of:

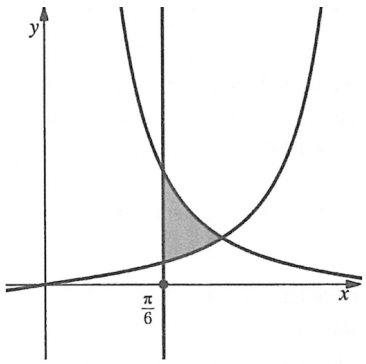
a) $\displaystyle\lim_{\delta x\to 0}\sum_{x=-\pi/2}^{\pi/2}\frac{5}{4}\sec^2\frac{5}{6}x\,\delta x$

b) $\displaystyle\lim_{\delta x\to 0}\sum_{x=\pi/9}^{2\pi/9}\frac{3}{8}\sec\frac{3}{2}x\tan\frac{3}{2}x\,\delta x$

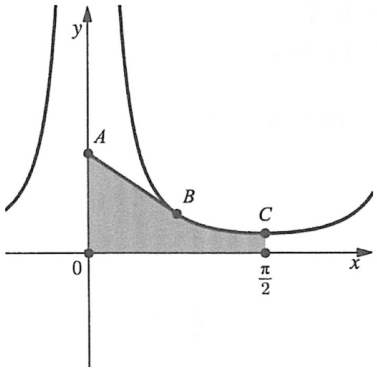
2.32 — Question 43. The diagram below shows the curve with equation $y = \sec^2 4x - 2$. Find the exact value of the shaded area.



2.32 — Question 44. The diagram below shows the curves $y = \operatorname{cosec} x \cot x$ and $y = \sec x \tan x$. Find the exact shaded area between the two curves.



2.32 — Question 45. The diagram below shows the curve with equation $y = \text{cosec}^2 x$. AB is the tangent to the curve at $x = \frac{\pi}{4}$. Find the exact shaded area, bounded by the curve, the tangent, the coordinate axes, and the line $x = \frac{\pi}{2}$.



2.33 — Question 1. Find

a) $\int (x+1)^3 \, dx$
b) $\int (x-4)^6 \, dx$
c) $\int 6(2+x)^7 \, dx$
d) $\int (10(x-1)^4 - 4(x-3)^5) \, dx$

2.33 — Question 2. Find the equation of the curve y given the gradient function is:

a) $\frac{dy}{dx} = (x-3)^{3/4}$

b) $\frac{dy}{dx} = (9+x)^{-3}$

c) $\frac{dy}{dx} = \sqrt{x+7}$

d) $\frac{dy}{dx} = \frac{4}{\sqrt{5+x}}$

2.33 — Question 3. The curve y has gradient function $\frac{dy}{dx} = \frac{6}{(x-2)^5}$ and passes through the point $(3, \frac{5}{2})$. Show that y can be written in the form $y = a + \frac{b}{c(x-2)^4}$ where a, b and c are integers.

2.33 — Question 4. The curve y has gradient function $\frac{dy}{dx} = \sqrt[3]{4+x}$ and passes through the point $(4, 3)$. Show that y can be written in the form $y = a(4+x)^b + c$, where a and b are rational numbers, and c is an integer.

2.33 — Question 5. Evaluate the following definite integrals:

a) $\int_2^3 (x-2)^8 \, dx$

b) $\int_{-6}^{-3} 24(6+x)^5 \, dx$

c) $\int_{-1}^{1} \sqrt{x+3} \, dx$

d) $\int_{-3}^{4} \frac{44}{\sqrt[3]{4+x}} \, dx$

2.33 — Question 6. Find the exact value of:

a) $\lim_{\delta x \to 0} \sum_{x=-5}^{26} (6+x)^{3/5} \delta x$

b) $\lim_{\delta x \to 0} \sum_{x=19}^{84} \frac{4}{\sqrt[4]{x-3}} \delta x$

2.33 — Question 7. Find

a) $\int (2x+1)^4 \, dx$

b) $\int (2-3x)^5 \, dx$

c) $\int 4(5+6x)^9 \, dx$

d) $\int (3(2x-3)^3 - 4(4-5x)^4) \, dx$

2.33 — Question 8. Find the equation of the curve y given the gradient function is:

a) $\frac{dy}{dx} = (5x + 9)^{7/3}$

b) $\frac{dy}{dx} = (6 - x)^{-5}$

c) $\frac{dy}{dx} = \sqrt{10 + 3x}$

d) $\frac{dy}{dx} = \frac{20}{\sqrt[3]{1-x}}$

2.33 — Question 9. The curve y has gradient function $\frac{dy}{dx} = \frac{6}{(3x-2)^2}$ and passes through the point $(0, 2)$. Show that y can be written in the form $y = \frac{ax+b}{cx+d}$, where a, b, c and d are integers.

2.33 — Question 10. The curve y has gradient function $\frac{dy}{dx} = \frac{14}{(4x+1)^{9/2}}$ and passes through the point $(0, 2)$. Show that y can be written in the form $y = a + \frac{b}{(4x+1)^c}$, where a and b are integers and c is a rational number.

2.33 — Question 11. Evaluate the following definite integrals:

a) $\int_2^3 (2x-5)^5 \, dx$

b) $\int_{-1/4}^{1/8} \sqrt{3+8x} \, dx$

c) $\int_1^2 \frac{15}{(5-6x)^2} \, dx$

d) $\int_{5/16}^{3/2} \frac{8}{\sqrt[3]{3+16x}} \, dx$

2.33 — Question 12. Find the exact value of:

a) $\lim_{\delta x \to 0} \sum_{x=-1/3}^{10} \frac{2}{5}(2+3x)^{3/5}\delta x$

b) $\lim_{\delta x \to 0} \sum_{x=1/9}^{1/3} \frac{243}{(4-9x)^4}\delta x$

2.33 — Question 13. Find

a) $\int 2x(x^2+3)^5\,dx$

b) $\int 12x^2(4x^3-5)^3\,dx$

c) $\int x(2x^2-5)^4\,dx$

d) $\int 2x^3\sqrt{x^4-3}\,dx$

2.33 — Question 14. Find the equation of the curve y given the gradient function is:

a) $\frac{dy}{dx} = 9x^2(3x^3+2)^{-2}$

b) $\frac{dy}{dx} = 10x^3(2-5x^4)^{5/2}$

c) $\frac{dy}{dx} = x\sqrt[3]{3+4x^2}$

d) $\frac{dy}{dx} = \frac{100x^8}{(2x^9+3)^6}$

2.33 — Question 15. The curve y has gradient function $\frac{dy}{dx} = 5x(5x^2-1)^7$ and passes through the point $(0,1)$. Show that y can be written in the form $y = a(5x^2-1)^b + c$, where a and c are rational numbers, and b is an integer.

2.33 — Question 16. The curve y has gradient function $\frac{dy}{dx} = \frac{9x^2}{\sqrt{27x^3-1}}$ and passes through the point $(\frac{1}{3}, 0)$. Show that y can be written in the form $y = a(27x^3 - 1)^b$, where a and b are rational numbers.

2.33 — Question 17. Evaluate the following definite integrals:

a) $\int_0^2 x(10x^2 + 1)^4 \, dx$

b) $\int_{1/\sqrt{2}}^{3/2} x(4x^2 - 1)^{2/3} \, dx$

c) $\int_2^3 \frac{6x^2}{(x^3+2)^2} \, dx$

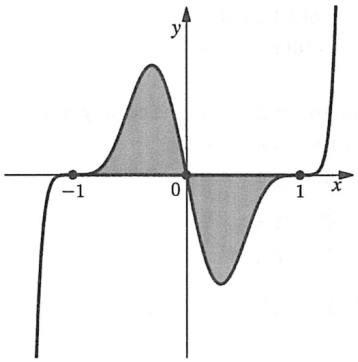
d) $\int_0^{1/2} \frac{800x^3}{(1-x^4)^3} \, dx$

2.33 — Question 18. Find the exact value of:

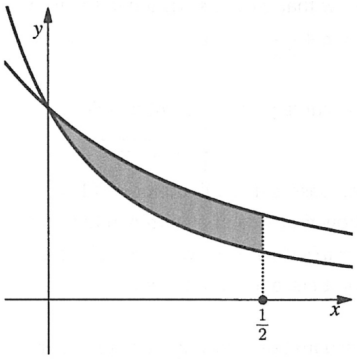
a) $\lim_{\delta x \to 0} \sum_{x=0}^{1} 98x(1+14x^2)^{5/2}\delta x$

b) $\lim_{\delta x \to 0} \sum_{x=0}^{9} \frac{\sqrt{\sqrt{x}+1}}{\sqrt{x}}\delta x$

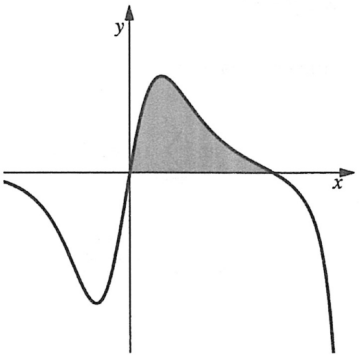
2.33 — Question 19. The diagram below shows the curve with equation $y = \frac{1}{2}x(x^2-1)^5$. Find the exact value of the shaded area.



2.33 — Question 20. The diagram below shows the curves $y = \frac{2}{(x+1)^2}$ and $y = \frac{2}{(2x+1)^2}$. Find the exact shaded area between the two curves.



2.33 — Question 21. The diagram below shows the curve with equation $y = \frac{12x-9x^2}{(2x^2-x^3+1)^3}$. Find the exact value of the shaded area.



2.34 — Question 1. $f'(x)e^{f(x)}$

Find

a) $\int e^{2x+1}\,dx$

b) $\int e^{5-3x}\,dx$

c) $\int xe^{x^2}\,dx$

d) $\int 4x^3e^{6x^4}\,dx$

2.34 — Question 2. Find

a) $\int \cos x\, e^{\sin x}\,dx$

b) $\int \sin 3x\, e^{6\cos 3x}\,dx$

c) $\int 4\cos\frac{1}{2}x\, e^{\sin\frac{1}{2}x}\,dx$

d) $\int 9\sin\frac{1}{3}x\, e^{3\cos\frac{1}{3}x}\,dx$

2.34 — Question 3. Find

a) $\displaystyle\int (\ln 2) 2^x e^{2^x}\, dx$

b) $\displaystyle\int 3^x e^{3^x}\, dx$

c) $\displaystyle\int \frac{1}{\ln 5} 5^x e^{5^x+1}\, dx$

d) $\displaystyle\int 4^x e^{4^{x+1}}\, dx$

2.34 — Question 4. Find the equation of the curve y given the gradient function is:

a) $\frac{dy}{dx} = \sec^2 x\, e^{\tan x}$

b) $\frac{dy}{dx} = 2 \operatorname{cosec} x \cot x\, e^{4 \operatorname{cosec} x}$

c) $\frac{dy}{dx} = 3 \sec 6x \tan 6x\, e^{\sec 6x}$

d) $\frac{dy}{dx} = -8 \operatorname{cosec}^2 \frac{1}{2} x\, e^{16 \cot \frac{1}{2} x}$

2.34 — Question 5. The curve y has gradient function

$$\frac{dy}{dx} = -\frac{e^{1/x}}{x^2}$$

and passes through the point $(1, -e)$. Write y in terms of x.

2.34 — Question 6. Evaluate the following definite integrals:

a) $\displaystyle\int_0^1 x^2 e^{x^3}\,dx$

b) $\displaystyle\int_0^{\pi/4} \cos 2x\, e^{\sin 2x}\,dx$

c) $\displaystyle\int_1^2 (\ln 8) 2^x e^{2^x}\,dx$

d) $\displaystyle\int_0^{\pi/16} 60\sec^2 4x\, e^{\tan 4x}\,dx$

2.34 — Question 7. Find the exact value of

$$\lim_{\delta x \to 0} \sum_{x=1}^{4} \frac{e^{2\sqrt{x}}}{\sqrt{x}} \delta x$$

2.34 — Question 8. $f'(x)\sin(f(x))$ **or** $f'(x)\cos(f(x))$

Find

a) $\displaystyle\int \sin\left(5x + \frac{\pi}{7}\right)\, dx$

b) $\displaystyle\int 8x \cos 2x^2\, dx$

c) $\displaystyle\int (2x+1)\sin(x^2 + x - 1)\, dx$

d) $\displaystyle\int (x^2 - 3)\cos\left(6x - \frac{2}{3}x^3\right)\, dx$

2.34 — Question 9. Find

a) $\int \cos x \sin(\sin x)\, dx$

b) $\int \sin x \sin(\cos x)\, dx$

c) $\int \cos x \cos(\sin x)\, dx$

d) $\int \sin x \cos(\cos x)\, dx$

2.34 — Question 10. Find

a) $\int (\ln 5) 5^x \sin(5^x)\, dx$

b) $\int 9^x \cos(9^x)\, dx$

c) $\int \dfrac{\sin(\ln x)}{x}\, dx$

d) $\int \dfrac{\cos(2 \ln x)}{x}\, dx$

2.34 — Question 11. Find the equation of the curve y given the gradient function is:

a) $\frac{dy}{dx} = \operatorname{cosec}^2 2x \sin(\cot 2x)$

b) $\frac{dy}{dx} = 4 \operatorname{cosec} 3x \cot 3x \cos(8 \operatorname{cosec} 3x)$

c) $\frac{dy}{dx} = 9 \sec^2 \frac{1}{3}x \sin(90 \tan \frac{1}{3}x)$

d) $\frac{dy}{dx} = \sec \frac{2}{3}x \tan \frac{2}{3}x \cos(18 \sec \frac{2}{3}x)$

2.34 — Question 12. The curve y has gradient function

$$\frac{dy}{dx} = 12x \sin x^2$$

and passes through the point $\left(\frac{\sqrt{\pi}}{2}, 24\right)$. Write y in terms of x.

2.34 — Question 13. Evaluate the following definite integrals:

a) $\displaystyle\int_{\ln(\pi/2)}^{\ln \pi} e^x \sin(e^x)\,dx$

b) $\displaystyle\int_{0}^{\sqrt{\pi}/4} 40x \cos 8x^2\,dx$

c) $\displaystyle\int_{\sqrt{2\pi}}^{\sqrt{3\pi}} 18x \sin x^2\,dx$

d) $\displaystyle\int_{\pi/4}^{\pi/3} (24x - 7\pi)\cos(12x^2 - 7\pi x + \pi^2)\,dx$

2.34 — Question 14. Find the exact value of

$$\lim_{\delta x \to 0} \sum_{x=-16/\pi}^{8/\pi} \frac{\sin(4/x)}{8x^2}\delta x$$

2.34 — Question 15. $\dfrac{f'(x)}{f(x)}$

Find

a) $\displaystyle\int \frac{1}{x+1}\,dx$

b) $\displaystyle\int \frac{2}{2x-5}\,dx$

c) $\displaystyle\int \frac{8}{4x+3}\,dx$

d) $\displaystyle\int \frac{9}{5x-7}\,dx$

e) $\displaystyle\int \frac{3}{1-4x}\,dx$

f) $\displaystyle\int \frac{15}{5-3x}\,dx$

2.34 — Question 16. Find

a) $\displaystyle\int \frac{2x}{x^2+3}\,dx$

b) $\displaystyle\int \frac{3x^2}{5+x^3}\,dx$

c) $\displaystyle\int \frac{15x}{5x^2-4}\,dx$

d) $\displaystyle\int \frac{10x^3}{6-x^4}\,dx$

2.34 — Question 17. Find

a) $\displaystyle\int \frac{2x+4}{x^2+4x-3}\,dx$

b) $\displaystyle\int \frac{3x^2-3}{x^3-3x+9}\,dx$

c) $\displaystyle\int \frac{x+1}{4x^2+8x+5}\,dx$

d) $\displaystyle\int \frac{6x^2-24x}{x^3-6x^2-1}\,dx$

2.34 — Question 18. Find

a) $\displaystyle\int \frac{e^x}{e^x+1}\,dx$

b) $\displaystyle\int \frac{(\ln 2)2^x-1}{2^x-x}\,dx$

c) $\displaystyle\int \tan x\,dx$

d) $\displaystyle\int \cot x\,dx$

2.34 — Question 19. Find

a) $\displaystyle\int \frac{\cos 2x}{\sin 2x + 1}\,dx$

b) $\displaystyle\int \frac{6\sin 3x}{4 - \cos 3x}\,dx$

c) $\displaystyle\int \frac{24\cos 6x}{3\sin 6x - 5}\,dx$

d) $\displaystyle\int \frac{18\sin 9x}{15\cos 9x + 4}\,dx$

2.34 — Question 20. Find the equation of the curve y given the gradient function is:

a) $\frac{dy}{dx} = \frac{(\ln 3)3^x + e^x}{3^x + e^x}$

b) $\frac{dy}{dx} = \frac{8 + 4\sec^2 2x}{4x + \tan 2x}$

c) $\frac{dy}{dx} = \frac{\cos x + \frac{1}{2\sqrt{x}}}{\sin x + \sqrt{x}}$

d) $\frac{dy}{dx} = \frac{30e^{3x} - 20e^{2x}}{e^{2x} - e^{3x}}$

2.34 — Question 21. The curve y has gradient function

$$\frac{dy}{dx} = \frac{2x+1}{(x-1)(x+2)}$$

and passes through the point $(2, 2\ln 2)$. Write y in terms of x.

2.34 — Question 22. Evaluate the following definite integrals:

a) $\displaystyle\int_0^1 \frac{12}{3x+1}\,dx$

b) $\displaystyle\int_{\sqrt{2}}^{\sqrt{3}} \frac{3x}{x^2-1}\,dx$

c) $\displaystyle\int_{\pi}^{2\pi} \frac{\sin x}{\cos x + 2}\,dx$

d) $\displaystyle\int_{\ln 2}^{\ln 4} \frac{4e^x}{4+e^x}\,dx$

2.34 — Question 23. Find the exact value of

$$\lim_{\delta x \to 0} \sum_{x=1}^{3} \frac{6x+1}{6x^2+2x+1} \delta x$$

2.34 — Question 24. $f'(x)(f(x))^n$

Find

a) $\int e^x (e^x+5)^3 \, dx$

b) $\int e^{3x} (e^{3x}-3)^4 \, dx$

c) $\int 2e^{4x} (4e^{4x}-1)^5 \, dx$

d) $\int 10e^{x/2} \sqrt{e^{x/2}+2} \, dx$

2.34 — Question 25. Find

a) $\int \cos x \sin^2 x \, dx$

b) $\int \cos x \sin^3 x \, dx$

c) $\int \sin x \cos^4 x \, dx$

d) $\int \sin 2x \cos^5 2x \, dx$

2.34 — Question 26. Find

a) $\int \sec^2 x(\tan x - 4)^5 \, dx$

b) $\int \sec^2 3x \tan^3 3x \, dx$

c) $\int 2^x(2^x + 4)^4 \, dx$

d) $\int 3^{x+1}(3^x - 2)^5 \, dx$

2.34 — Question 27. Find the equation of the curve y given the gradient function is:

a) $\frac{dy}{dx} = \frac{(\ln x+3)^3}{x}$

b) $\frac{dy}{dx} = \frac{(2\ln x+5)^5}{12x}$

c) $\frac{dy}{dx} = \sec x \tan x \,(\sec x + 2)^3$

d) $\frac{dy}{dx} = \operatorname{cosec}^8 x \cot x$

2.34 — Question 28. Evaluate the following definite integrals:

a) $\displaystyle\int_{\ln 2}^{\ln 4} e^x(4e^x - 5)^3 \, dx$

b) $\displaystyle\int_{0}^{\pi/2} \cos x \sin^4 x \, dx$

c) $\displaystyle\int_{2\pi}^{3\pi} \sin \frac{1}{2}x \cos^5 \frac{1}{2}x \, dx$

d) $\displaystyle\int_{e}^{e^2} \frac{(\ln x + 3)^5}{2x} \, dx$

2.34 — Question 29. Mix
Given that k is a positive constant, find

$$\int_{k}^{2k} \frac{8}{4x+k}\,dx$$

2.34 — Question 30. Given that k is a positive constant, find

$$\int_{1/k}^{2/k} \frac{k}{kx+5}\,dx$$

2.34 — Question 31. Given that k is a positive constant, find

$$\int_{2/k}^{3/k} \frac{k^2 x}{k^2 x^2 - 2}\, dx$$

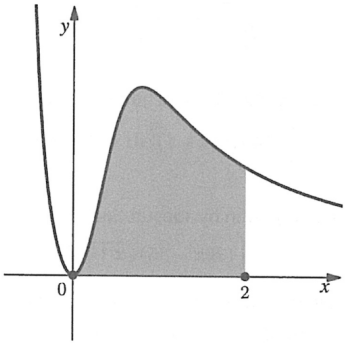
2.34 — Question 32. Given that k is a positive constant, show that

$$\int_{\sqrt{k}}^{\sqrt{4k}} 8xe^{x^2}\, dx = 4e^k(e^{3k} - 1)$$

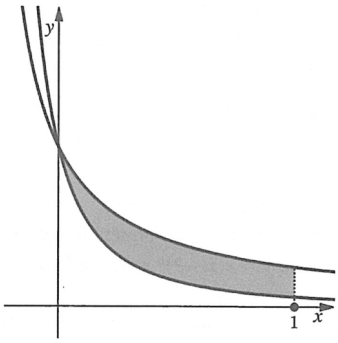
2.34 — Question 33. Find the exact value of

$$\int_{\ln(\sqrt{\pi}/4)}^{\ln(\sqrt{\pi}/2)} 8e^{2x}\sin(4e^{2x})\,dx$$

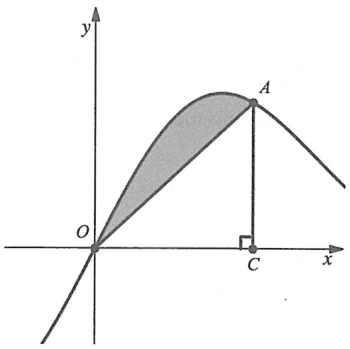
2.34 — Question 34. The diagram below shows the curve with equation $y = \dfrac{2x^2}{4x^3+1}$. Find the exact value of the shaded area.



2.34 — Question 35. The diagram below shows the curves $y = \dfrac{3}{3x+1}$ and $y = \dfrac{3}{(3x+1)^2}$. Find the exact shaded area between the two curves.

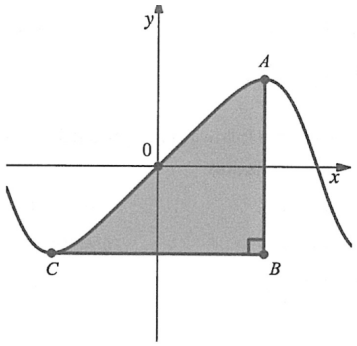


2.34 — Question 36. The diagram below shows the curve with equation $y = \sin 2x \cos^5 2x$ where C has x-coordinate $\dfrac{\pi}{12}$. Find the exact shaded area.



2.34 — Question 37. Prerequisite knowledge for Q37: Quotient Rule

The diagram below shows $y = \dfrac{\sin 3x}{\cos 3x + 2}$. A and C are turning points on the curve, and the angle ABC is 90°. Find the exact shaded area.



2.35 — Question 1. Each of the following integrals can be found using reversing the chain rule. However, in each case, evaluate the integrals using the given substitution:

a) $\int (3x+1)^4 \, \mathrm{d}x$ $u = 3x + 1$

b) $\int \frac{4}{(2x-3)^3} \, \mathrm{d}x$ $u = 2x - 3$

c) $\int \sqrt{8x+5} \, \mathrm{d}x$ $u = 8x + 5$

d) $\int \frac{6}{\sqrt[3]{5-2x}} \, \mathrm{d}x$ $u = 5 - 2x$

2.35 — Question 2. It is given that $y = x(x-3)^5$

a) Describe the transformation that would map $y = x(x-3)^5$ onto the image $y = (x+3)x^5$

b) Use the substitution $u = x - 3$ to show that

$$\int x(x-3)^5 \, \mathrm{d}x = \int (u+3)u^5 \, \mathrm{d}u$$

2.35 — Question 3. It is given that $y = 2x\sqrt{x+4}$

a) Describe the transformation that would map $y = 2x\sqrt{x+4}$ onto the image $y = 2(x-4)\sqrt{x}$

b) Use the substitution $u = x + 4$ to show that

$$\int 2x\sqrt{x+4} \, \mathrm{d}x = \int 2(u-4)\sqrt{u} \, \mathrm{d}u$$

2.35 — Question 4. Evaluate the following integrals using the given substitution:

a) $\int x\sqrt{x+1}\,\mathrm{d}x$ $u = x+1$

b) $\int 2x\sqrt{2x+3}\,\mathrm{d}x$ $u = 2x+3$

c) $\int x^3\sqrt{x^2+1}\,\mathrm{d}x$ $u = x^2+1$

2.35 — Question 5. Evaluate the following integrals using the given substitution:

a) $\int \frac{x}{x+1}\,\mathrm{d}x$ $u = x+1$

b) $\int \frac{2x}{2x+3}\,\mathrm{d}x$ $u = 2x+3$

c) $\int \frac{x^3}{x^2+1}\,\mathrm{d}x$ $u = x^2+1$

2.35 — Question 6. Evaluate the following integrals using the given substitution:

a) $\int \frac{x^2}{x+1}\,\mathrm{d}x$ $u = x + 1$

b) $\int \frac{4x^2}{2x+3}\,\mathrm{d}x$ $u = 2x + 3$

c) $\int \frac{x^5}{x^2+1}\,\mathrm{d}x$ $u = x^2 + 1$

2.35 — Question 7. Evaluate the following integrals using the given substitution:

a) $\int x^2\sqrt{x+1}\,\mathrm{d}x$ $u = x + 1$

b) $\int 4x^2\sqrt{2x+3}\,\mathrm{d}x$ $u = 2x + 3$

c) $\int x^3\sqrt{x^2+1}\,\mathrm{d}x$ $u = x^2 + 1$

2.35 — Question 8. Use the substitution $u = 2 - 3x$ to find

$$\int 189x(2-3x)^5 \, \mathrm{d}x$$

2.35 — Question 9. Use the substitution $u = x + 1$ to find the exact value of

$$\int_0^1 3x(x+1)^{-\frac{1}{2}} \, \mathrm{d}x$$

2.35 — Question 10. Use the substitution $u = 3x + 4$ to show that

$$\int_{-2}^{-\frac{4}{3}} (6x+1)(3x+4)^7 \, \mathrm{d}x = \frac{3040}{27}$$

2.35 — Question 11. Use integration by substitution to show that

$$\int_{1}^{2} \frac{x}{x+2} \, \mathrm{d}x = a + \ln \frac{b}{c}$$

where a, b and c are positive integers.

2.35 — Question 12. Use integration by substitution to show that

$$\int_1^3 \frac{9x}{3x+5}\,\mathrm{d}x = 6 + 5\ln\frac{4}{7}$$

2.35 — Question 13. Use the substitution $u = 2x + 3$ to find the exact value of

$$\int_1^2 \frac{8x-1}{(2x+3)^3}\,\mathrm{d}x$$

2.35 — Question 14. Use integration by substitution to show that

$$\int_{\frac{3}{2}}^{\frac{19}{2}} 5x\sqrt{2x-3}\,\mathrm{d}x = 672$$

2.35 — Question 15. Use integration by substitution to show that

$$\int_{0}^{1} 140x(4x+1)^{\frac{3}{2}}\,\mathrm{d}x = 1 + 225\sqrt{5}$$

2.35 — Question 16. Use integration by substitution to find

$$\int (30x - 45)\sqrt{x+2}\,\mathrm{d}x$$

in the form $(x+2)^{\frac{3}{2}}(ax+b)+c$, where a and b are integers, and c is a constant.

2.35 — Question 17. Use integration by substitution to find the exact value of

$$\int_{\frac{1}{12}}^{\frac{3}{4}} (2-6x)(12x-1)^{\frac{1}{3}}\,\mathrm{d}x$$

2.35 — Question 18. Use the substitution $u = x^2 + 5$ to find

$$\int \frac{2x^3 - x}{\sqrt{x^2 + 5}} \, \mathrm{d}x$$

2.35 — Question 19. Use the substitution $u = 4 - x^4$ to find the exact value of

$$\int_0^1 \frac{x^7}{4 - x^4} \, \mathrm{d}x$$

2.35 — Question 20. Use the substitution $u = x^5 + 3$ to show that

$$\int_0^1 \frac{20x^9}{(x^5+3)^2}\,\mathrm{d}x = \ln\frac{a}{b} - 1$$

where a and b are positive integers.

2.35 — Question 21. Use the substitution $u = 2\sqrt{x} - 1$ to find

$$\int \frac{2}{2\sqrt{x}-1}\,\mathrm{d}x$$

2.35 — Question 22. Evaluate the following definite integral

$$\int_1^4 \frac{2}{\sqrt{x}+1}\,\mathrm{d}x$$

using the substitution $u = \sqrt{x}+1$

2.35 — Question 23. Evaluate the following definite integral

$$\int_2^5 \frac{2x^2}{\sqrt{x-1}}\,\mathrm{d}x$$

using the substitution $u = \sqrt{x-1}$

2.35 — Question 24. Use integration by substitution to find

$$\int \frac{3x}{1+\sqrt{x}}\,\mathrm{d}x$$

2.35 — Question 25. Evaluate the following definite integral

$$\int_1^4 \frac{1}{\sqrt{x}(2+\sqrt{x})}\,\mathrm{d}x$$

using the substitution $u = \sqrt{x}$

2.35 — Question 26. Use the substitution $u = 1 + 2\sqrt{x}$ to show that

$$\int_1^9 \frac{1+\sqrt{x}}{1+2\sqrt{x}}\,\mathrm{d}x$$

can be written in the form $a + b\ln c$, where a is an integer, and b and c are rational numbers.

2.35 — Question 27. Evaluate the following definite integral

$$\int_0^3 \frac{8-x}{3+\sqrt{x+1}}\,\mathrm{d}x$$

using the substitution $u = \sqrt{x+1}$

2.35 — Question 28. Evaluate the following definite integrals using the given substitutions:

a) $\int_0^{16} \frac{4}{1+\sqrt{x}}\,\mathrm{d}x$ $u^2 = x$

b) $\int_1^3 \frac{8x}{\sqrt{4x-3}}\,\mathrm{d}x$ $u^2 = 4x - 3$

2.35 — Question 29. Evaluate the following definite integral:

$$\int_1^4 \frac{12}{4\sqrt{x}+3x}\,\mathrm{d}x$$

a) Using the substitution $u = \sqrt{x}$

b) Using the substitution $u = 3\sqrt{x} + 4$

2.35 — Question 30. Use the substitution $u^2 = 2x^2 + 1$ to evaluate

$$\lim_{\delta x \to 0} \sum_{x=0}^{2} \frac{x^3}{\sqrt{2x^2+1}} \delta x$$

2.35 — Question 31. Standard Functions Each of the following integrals can be found using reversing the chain rule. However, in each case, evaluate the integrals using the given substitution:

a) $\int \cos x \sin^4 x \, \mathrm{d}x$ $u = \sin x$

b) $\int 8xe^{x^2+1} \, \mathrm{d}x$ $u = x^2 + 1$

c) $\int \sin x \, \mathrm{e}^{\cos x} \, \mathrm{d}x$ $u = \cos x$

d) $\int \frac{\sec^2 x}{\tan x} \, \mathrm{d}x$ $u = \tan x$

2.35 — Question 32. Evaluate the following definite integral

$$\int_0^{\frac{\pi}{2}} \frac{6\cos x}{9+\sin x}\,\mathrm{d}x$$

using the substitution $u = 9 + \sin x$

2.35 — Question 33. Use integration by substitution to find

$$\int \frac{5\mathrm{e}^x}{5+\mathrm{e}^x}\,\mathrm{d}x$$

2.35 — Question 34. Evaluate the following definite integrals using the given substitutions:

a) $\int_0^{\ln 2} \frac{3e^{2x}}{3+e^x}\,dx$ $u = 3 + e^x$

b) $\int_{\ln 2}^{\ln 3} \frac{8e^x}{(2+e^x)^2}\,dx$ $u = 2 + e^x$

2.35 — Question 35. Evaluate the following definite integrals using the given substitutions:

a) $\int_0^{\frac{\pi}{4}} 8\cos 2x \sin^5 2x\,dx$ $u = \sin 2x$

b) $\int_0^{\frac{\pi}{8}} \sin^2 2x \cos^3 2x\,dx$ $u = \sin 2x$

2.35 — Question 36. Evaluate the following definite integrals using the given substitutions:

a) $\int_0^{\frac{\pi}{4}} \sec^2 x \, e^{\tan x + 1} \, dx$ $u = \tan x + 1$

b) $\int_0^{\frac{\pi}{4}} \sec^4 x \sqrt[3]{\tan x} \, dx$ $u = \tan x$

2.35 — Question 37. Use the substitution $u = 1 + \tan x$ to find

$$\int \frac{1}{\cos^2 x (1 + \tan x)^2} \, dx$$

2.35 — Question 38. Evaluate the following definite integral

$$\int_1^2 \frac{5^x}{(5^x-1)^3}\,\mathrm{d}x$$

a) Using the substitution $u = 5^x - 1$
b) Using the substitution $u = 5^x$

2.35 — Question 39. Evaluate the following definite integral

$$\int_0^1 8^x(8^x+1)^{\frac{3}{2}}\,\mathrm{d}x$$

a) Using the substitution $u = 8^x + 1$
b) Using the substitution $u = 8^x$

2.35 — Question 40. Use the substitution $u = 3 - \ln x$ to find

$$\int \frac{6}{x(3 - \ln x)^5}\,\mathrm{d}x$$

2.35 — Question 41. Evaluate the following definite integrals using the given substitutions:

a) $\int_1^{2\mathrm{e}} \frac{4}{x(5+\ln x)^3}\,\mathrm{d}x$ $u = 5 + \ln x$

b) $\int_1^{\mathrm{e}} \frac{2\ln x}{x(4-\ln x)^2}\,\mathrm{d}x$ $u = 4 - \ln x$

2.35 — Question 42. Trigonometric and other substitutions Use the substitution $x = \sin u$ to find:

a) $\int \frac{1}{\sqrt{1-x^2}}\,\mathrm{d}x$

b) $\int \frac{1}{(1-x^2)^{\frac{3}{2}}}\,\mathrm{d}x$

2.35 — Question 43. Evaluate the following definite integral

$$\int_0^{\frac{1}{4}} \frac{1}{\sqrt{1-4x^2}}\,\mathrm{d}x$$

2.35 — Question 44. Use the substitution $x = \frac{3}{2}\sin u$ to find

$$\int \frac{1}{\sqrt{9-4x^2}}\,\mathrm{d}x$$

2.35 — Question 45. Use the given substitutions to find:

a) $\int \frac{1}{1+x^2}\,\mathrm{d}x$ $x = \tan u$

b) $\int \frac{5}{25+x^2}\,\mathrm{d}x$ $x = 5\tan u$

2.35 — Question 46. Evaluate the following definite integral

$$\int_0^1 \frac{1}{(1+x^2)^2}\,\mathrm{d}x$$

using the substitution $x = \tan u$

2.35 — Question 47. Evaluate the following definite integral

$$\int_0^1 \frac{1-x^2}{1+x^2}\,\mathrm{d}x$$

using the substitution $x = \tan u$

2.35 — Question 48. Evaluate the following definite integrals using the given substitutions:

a) $\int_2^3 \frac{1}{(x-2)^2+1}\,\mathrm{d}x$ $x = 2 + \tan u$

b) $\int_{-3}^{-1} \frac{1}{(x+3)^2+4}\,\mathrm{d}x$ $x = 2\tan u - 3$

2.35 — Question 49. Use integration by substitution to find

$$\int_5^8 \frac{1}{x^2 - 10x + 34}\,\mathrm{d}x$$

2.35 — Question 50. Use integration by substitution to find

$$\int_5^8 \frac{12x}{x^2 - 10x + 34}\,\mathrm{d}x$$

2.35 — Question 51. Evaluate the following definite integral

$$\int_{\frac{2\sqrt{3}}{3}}^2 \frac{6}{x^2\sqrt{x^2 - 1}}\,\mathrm{d}x$$

using the substitution $x = \operatorname{cosec} u$

2.35 — Question 52. Given that $x > 3$, use the substitution $x = 3\operatorname{cosec} u$ to show that

$$\int \frac{1}{x^2\sqrt{x^2-9}}\,\mathrm{d}x = \frac{\sqrt{x^2-9}}{9x} + c$$

2.35 — Question 53. Prerequisite knowledge for Q53: Integration by Parts Use the substitution $2x = \mathrm{e}^u$ to find

$$\int (\ln 2x)^2\,\mathrm{d}x$$

2.35 — Question 54. Use the substitution $x = \sin u$ to find

$$\int \frac{3}{(1-x^2)^{\frac{5}{2}}}\,\mathrm{d}x$$

2.35 — Question 55. Prerequisite knowledge for Q55–Q58: Double Angle Formulae Use the substitution $u = \cos x - 2$ to find

$$\int \frac{4\sin 2x}{\cos x - 2}\,\mathrm{d}x$$

2.35 — Question 56. Use integration by substitution to find

$$\int \frac{x^2}{\sqrt{1-x^2}}\,\mathrm{d}x$$

2.35 — Question 57. Evaluate the following definite integral using the given substitution:

a) $\int_0^{\frac{3}{2}} \sqrt{9-x^2}\,\mathrm{d}x$ $x = 3\sin u$

b) $\int_0^{\frac{1}{3}} \sqrt{1-9x^2}\,\mathrm{d}x$ $x = \frac{1}{3}\sin u$

2.35 — Question 58. Evaluate the following definite integrals using the given substitutions:

a) $\int_0^1 \frac{1-x^2}{(1+x^2)^2}\,\mathrm{d}x$ $x = \tan u$

b) $\int_0^{\frac{5}{2}} \sqrt{\frac{x}{5-x}}\,\mathrm{d}x$ $x = 5\sin^2 u$

2.35 — Question 59. Prerequisite knowledge for Q59–Q62: Integration by Parts Use integration by substitution to find:

a) $\int 2x^3 \ln(x^2+4)\,\mathrm{d}x$

b) $\int 3^x \ln(3^x+1)\,\mathrm{d}x$

2.35 — Question 60. Use the substitution $u = \sqrt{x+2}$ to find

$$\int e^{4\sqrt{x+2}}\,dx$$

2.35 — Question 61. Evaluate the following definite integral

$$\int_0^4 e^{\sqrt{2x+1}}\,dx$$

using the substitution $u^2 = 2x + 1$

2.35 — Question 62. Evaluate the following definite integrals using the given substitutions:

a) $\int_0^1 5x^5 \ln(10x^3+1)\,\mathrm{d}x$ $u = 10x^3+1$

b) $\int_{-1}^{\frac{\pi^2}{4}-1} \cos(\sqrt{x+1})\,\mathrm{d}x$ $u = \sqrt{x+1}$

2.35 — Question 63. Prerequisite knowledge for Q63–Q65: Partial Fractions Evaluate the following definite integrals

$$\int_0^{\ln 5} \frac{10}{5+\mathrm{e}^x}\,\mathrm{d}x$$

using the substitution $u = \mathrm{e}^x$

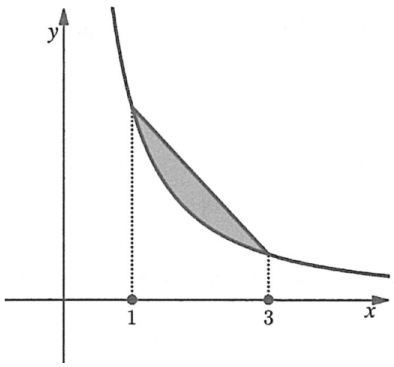
2.35 — Question 64. Evaluate the following definite integrals

$$\int_1^4 \frac{2}{x(1+3\sqrt{x})}\,\mathrm{d}x$$

using the substitution $x = u^2$

2.35 — Question 65. The diagram below shows the curve with equation

$$y = \frac{1}{x\sqrt{x+1}}$$



a) Using the substitution $u = \sqrt{x+1}$, find the exact value of the definite integral

$$\int_1^3 \frac{1}{x\sqrt{x+1}}\,\mathrm{d}x$$

b) Find the exact value of the shaded area.

2.35 — Question 66. Prerequisite knowledge for Q66: Implicit Differentiation

a) Use the substitution $\mathrm{e}^x = \sec^2 u$ to show that:

$$\int \sqrt{\mathrm{e}^x - 1}\,\mathrm{d}x = \int 2\tan^2 u\,\mathrm{d}u$$

b) Hence find the exact value of:

$$\int_0^{\ln \frac{4}{3}} \sqrt{\mathrm{e}^x - 1}\,\mathrm{d}x$$

2.36 — Question 1. Find

a) $\int xe^x\,dx$

b) $\int (x-1)e^x\,dx$

c) $\int (2x+5)e^x\,dx$

2.36 — Question 2. Evaluate the following definite integral:

$$\int_{-\frac{1}{3}}^{\frac{1}{3}} xe^{3x}\,dx$$

2.36 — Question 3. Evaluate the following definite integral:

$$\int_{-1}^{0} \frac{x+1}{3e^{x}}\,dx$$

2.36 — Question 4. The curve y has gradient function

$$\frac{dy}{dx} = 2x\sqrt{e^x}$$

and passes through the point $(\ln 4, 2)$. Show that y can be written in the form

$$y = a(x-2)\sqrt{e^x} + b\ln 4 + c,$$

where a, b and c are integers.

2.36 — Question 5. Find

a) $\displaystyle\int x\sin x\,dx$

b) $\displaystyle\int \left(x+\frac{\pi}{3}\right)\sin x\,dx$

2.36 — Question 6. Evaluate the following definite integral:

$$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} x \sin 2x \, dx$$

2.36 — Question 7. Find

a) $\int x \cos x \, dx$

b) $\int \left(x - \frac{\pi}{6}\right) \cos x \, dx$

2.36 — Question 8. Evaluate the following definite integral:

$$\int_{\frac{3\pi}{2}}^{\frac{5\pi}{2}} \left(x - \frac{2\pi}{3}\right) \cos x \, dx$$

2.36 — Question 9. Evaluate the following definite integral:

$$\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \left(2x - \frac{\pi}{4}\right) \sin 2x \, dx$$

2.36 — Question 10. Evaluate the following definite integral:

$$\int_{-\frac{\pi}{8}}^{\frac{\pi}{8}} \left(x + \frac{\pi}{8}\right) \cos 4x \, dx$$

2.36 — Question 11. The curve y has gradient function

$$\frac{dy}{dx} = 64x \sin 8x$$

and passes through the point $\left(\frac{\pi}{16}, 1\right)$. Show that y can be written in the form

$$y = a \sin 8x + bx \cos 8x,$$

where a and b are integers.

2.36 — Question 12. The curve y has gradient function

$$\frac{dy}{dx} = 6x\cos\frac{1}{3}x$$

and passes through the point $(\pi, -1)$. Show that y can be written in the form

$$y = ax\sin\frac{1}{3}x + b\cos\frac{1}{3}x + p\sqrt{3}\pi + q,$$

where a, b, p and q are integers.

2.36 — Question 13. Find

a) $\displaystyle\int x\ln x\,dx$

b) $\displaystyle\int x^2\ln x\,dx$

c) $\displaystyle\int (x+4)\ln x\,dx$

d) $\displaystyle\int (x^3+3x)\ln x\,dx$

2.36 — Question 14. Evaluate the following definite integral:

$$\int_1^2 \frac{\ln 8x}{x^4}\,dx$$

2.36 — Question 15. The curve y has gradient function

$$\frac{dy}{dx} = 4x \ln \frac{1}{2}x$$

and passes through the point $(2e, 1)$. Show that y can be written in the form

$$y = ax^2 + bx^2 \ln \frac{1}{2}x + pe^2 + q,$$

where a, b, p and q are integers.

2.36 — Question 16. Find

a) $\int x2^x \, dx$

b) $\int (3x+9)3^x \, dx$

2.36 — Question 17. Evaluate the following definite integral:

$$\int_0^1 (x+2)2^x \, dx$$

2.36 — Question 18. Evaluate the following definite integral:

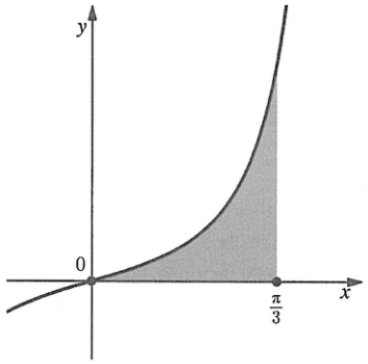
$$\int_1^2 10x5^{x-1}\,dx$$

2.36 — Question 19. The curve y has gradient function

$$\frac{dy}{dx} = \frac{x}{4^x}$$

and passes through the point $(\ln 4, 1)$. Write y in terms of x.

2.36 — Question 20. The diagram below shows the curve $y = x\sec^2 x$. Show that the exact shaded area is $\frac{\pi}{\sqrt{3}} - \ln 2$.



2.36 — Question 21. Evaluate the following definite integral:

$$\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} 18x\,\mathrm{cosec}^2 x\,dx$$

2.36 — Question 22. Find

a) $\int \ln x \, dx$

b) $\int (\ln x)^2 \, dx$

2.36 — Question 23. Prerequisite knowledge for Q23: Integration by Substitution

Evaluate the following definite integral:

$$\int_3^{e^2+2} \ln(x-2) \, dx$$

2.36 — Question 24. Integration by Parts Twice Find

a) $\int x^2 e^x \, dx$

b) $\int \frac{x^2}{e^x} \, dx$

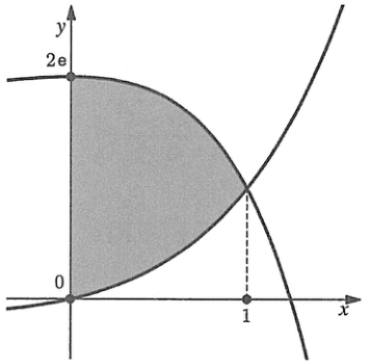
c) $\int x^2 \sin x \, dx$

d) $\int x^2 2^x \, dx$

2.36 — Question 25. Evaluate the following definite integral:

$$\int_0^{\frac{1}{4}} 20x^2 e^{-4x} \, dx$$

2.36 — Question 26. The diagram below shows the curves $y = xe^x$ and $y = 2e - x^2e^x$. Find the exact shaded area between the two curves.



2.36 — Question 27. Evaluate the following definite integral:

$$\int_1^e (\ln x)^3 \, dx$$

2.36 — Question 28. Cyclic Integration Find

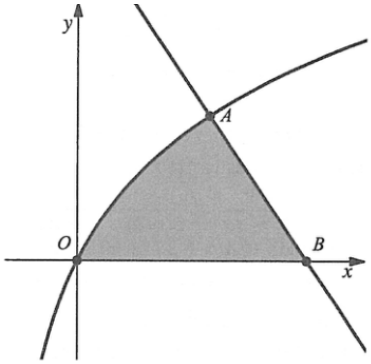
a) $\int e^x \sin x \, dx$

b) $\int e^x \cos x \, dx$

2.36 — Question 29. Evaluate the following definite integral:

$$\int_0^{\frac{\pi}{2}} e^{5x} \sin 2x \, dx$$

2.36 — Question 30. The diagram below shows the curve $y = \ln(2x - k)$. It is given that the curve passes through the origin, O, and the point A with x-coordinate 1. The normal to the curve at A intersects the x-axis at B. Find the exact shaded area of the shape OAB.



2.38 — Question 1. It is given that $f(x) = \dfrac{8x + 23}{x + 2}$.

a) Write $f(x)$ in the form $A + \dfrac{B}{x + 2}$.

b) Hence find $\int f(x)\,dx$.

2.38 — Question 2. It is given that $f(x) = \dfrac{2x^2 - 2x - 1}{x + 1}$.

a) Write $f(x)$ in the form $Ax + B + \dfrac{C}{x+1}$.

b) Hence find $\int f(x)\,dx$.

2.38 — Question 3.

a) Express $\dfrac{4}{(x-3)(x+1)}$ as partial fractions.

b) Show that $\displaystyle\int \frac{4}{(x-3)(x+1)}\,dx = \ln\left|\frac{x-3}{x+1}\right| + c$.

2.38 — Question 4.

a) Find $\displaystyle\int \frac{10}{25-x^2}\,dx$.

b) Hence show that $\displaystyle\int_{-1}^{1} \frac{10}{25-x^2}\,dx = \ln\frac{a}{b}$ where a and b are integers to be found.

2.38 — Question 5.

a) Find $\displaystyle\int \frac{2x-1}{(x+5)(x-6)}\,dx$.

b) Hence find the value of k such that $\displaystyle\int_{7}^{k} \frac{2x-1}{(x+5)(x-6)}\,dx = \ln 5$.

2.38 — Question 6.

a) Express $\dfrac{k-4x}{x(2-x)}$ as partial fractions, where k is a positive constant.

b) Find the value of k such that $\displaystyle\int_3^4 \frac{k-4x}{x(2-x)}\,dx = \ln\frac{128}{27}$.

2.38 — Question 7.

a) Express $\dfrac{23x+11}{(3-x)(x+2)(3x-1)}$ as partial fractions.

b) Write $\displaystyle\int_1^2 \frac{23x+11}{(3-x)(x+2)(3x-1)}\,dx$ in the form $\ln k$, where k is a positive constant.

2.38 — Question 8.

a) Express $\dfrac{4x^2+1}{x^2(2x+1)}$ as partial fractions.

b) Show that $\displaystyle\int \frac{4x^2+1}{x^2(2x+1)}\,dx = \frac{a}{x} + \ln f(x) + c$ where a is an integer to be found.

2.38 — Question 9.

a) Write $\dfrac{4x-2}{(x-2)^2}$ in the form $\dfrac{A}{x-2} + \dfrac{B}{(x-2)^2}$ where A and B are constants.

b) Show that $\displaystyle\int_3^4 \frac{4x-2}{(x-2)^2}\,dx = a + \ln b$ where a and b are integers to be found.

2.38 — Question 10.

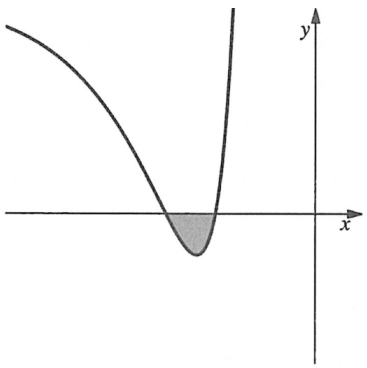
a) Find $\dfrac{10x^2+6}{(x+1)^2(3x-1)}$ as partial fractions.

b) Hence show that $\displaystyle\int_1^{5/3} \frac{10x^2+6}{(x+1)^2(3x-1)}\,dx = a + b\ln 2 + c\ln 9$ where a, b and c are constants to be found.

2.38 — Question 11. The diagram below shows the curve with equation

$$y = \frac{x^2+5x+6}{x^2(2-x)}, \qquad x \neq 0,\ x \neq 2.$$

Find the exact value of the shaded area.



2.38 — Question 12. Find the exact value of

$$\lim_{\delta x \to 0} \sum_{x=0}^{1/8} \frac{360x^2 + 157x + 16}{(3x+1)(2-5x)(8x+3)}\,\delta x.$$

2.38 — Question 13. Prerequisite knowledge for Q13–Q14: Integration by Substitution Use the substitution $u = \sqrt{x} - 2$ to find

$$\int \frac{9}{2(\sqrt{x}-2)(\sqrt{x}+1)(\sqrt{x}+4)}\,dx.$$

2.38 — Question 14. Use the substitution $u = \sqrt{x-2}$ to show that

$$\int_3^6 \frac{6}{(3+2\sqrt{x-2})(x-2)}\,dx = 4\ln\frac{10}{7}.$$

2.39 — Question 1.

a) Write $\cos 2x$ in terms of $\cos x$

b) Hence show that $\int_0^{\pi/4} \cos^2 x\,dx = \frac{2+\pi}{8}$

2.39 — Question 2.

a) Write $\cos 2x$ in terms of $\sin x$

b) Hence show that $\int_0^{\pi/2} \sin^2 x \, dx = \frac{\pi}{4}$

2.39 — Question 3. Find

a) $\int (1 + \sin x)^2 \, dx$

b) $\int (1 - \cos x)^2 \, dx$

c) $\int (\sin x - \cos x)^2 \, dx$

d) $\int (\sin x - \cos x)(\sin x + \cos x) \, dx$

2.39 — Question 4. Find

a) $\int \sin x \cos x \, dx$

b) $\int 8 \sin 2x \cos 2x \, dx$

c) $\int 12 \sin 5x \cos 5x \, dx$

d) $\int \cos^2 6x \tan 6x \, dx$

2.39 — Question 5. Evaluate the following definite integrals:

a) $\int_0^{\pi/2} (4 - 3\sin x)^2 \, dx$

b) $\int_0^{\pi/2} (1 - 6\cos x)^2 \, dx$

2.39 — Question 6. Evaluate the following definite integrals:

a) $\int_0^{\pi/4}(1-8\sin 2x)^2\,dx$

b) $\int_{-\pi}^{\pi}(7+9\cos 3x)^2\,dx$

2.39 — Question 7. Find

a) $\int(2\cos^2 3x-1)\,dx$

b) $\int(1-2\sin^2 5x)\,dx$

c) $\int(\sin 4x\cot 2x-1)\,dx$

d) $\int(1-\sin 20x\tan 10x)\,dx$

2.39 — Question 8. Find $\displaystyle\int \frac{10\sin x\cos x}{2\cos^2 x+3}\,dx$

2.39 — Question 9. The curve y has gradient function

$$\frac{dy}{dx}=\frac{4\tan x}{\sqrt{2\cos 2x+2}}$$

and passes through the point $(\pi,6)$.
Show that y can be written in the form $y=a\sec x+b$, where a and b are integers to be found.

2.39 — Question 10. Evaluate the following definite integral

$$\int_0^{\pi/4} \frac{\sqrt{1-\cos 2x}}{\cos 2x + 1}\, dx$$

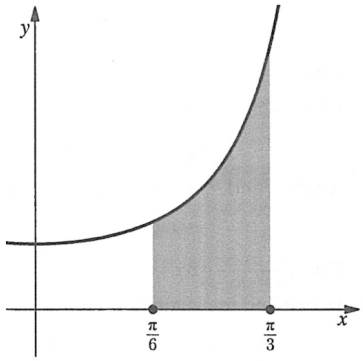
2.39 — Question 11. Find the exact value of

$$\lim_{\delta x \to 0} \sum_{x=\pi/4}^{\pi/2} \frac{(\cos 2x + 1)\operatorname{cosec} x}{\sin 2x}\, \delta x$$

2.39 — Question 12. The diagram below shows the curve with equation

$$y = \frac{2\cos x}{\sin 2x(\operatorname{cosec} x - \sin x)}$$

Find the exact value of the shaded area.



2.39 — Question 13.

a) Use the Compound Angle Formulae to show that $\sin 3x \equiv 3\sin x - 4\sin^3 x$

b) Hence find $\int \sin^3 x \, dx$

2.39 — Question 14.

a) Use the Compound Angle Formulae to show that $\cos 3x \equiv 4\cos^3 x - 3\cos x$

b) Hence find $\int \cos^3 x \, dx$

2.39 — Question 15. Prerequisite knowledge for Q15-Q16: Integration by Substitution Evaluate the following definite integral

$$\int_0^{1/2} \sqrt{\frac{16x}{1-x}} \, dx$$

using the substitution $x = \cos^2 u$

2.39 — Question 16. Evaluate the following definite integral

$$\int_{-1}^{5} \sqrt{(x+1)(5-x)}\, dx$$

using the substitution $x = 2 - 3\cos u$